Last time... Defined (real) vector spaces. 10 properties (axioms)

- Examples $\mathbb{R}^n$, $M_{m\times n}$, $\mathbb{P}_n$, $\mathbb{P}$, real valued functions on $\mathbb{R}$

These are the "parent" vector spaces

Additional properties of a vector space V

(i) $0u = 0$ $\forall u \in V$, (ii) $c\vec{0} = \vec{0}$ $\forall c \in \mathbb{R}$

zero in $\mathbb{R}$ zero in V zero in V zero in V

(iii) $(-1)u = -u$ $\forall u \in V$. $u+(-u)=0$

Def Suppose V is a (real) vector space, and $H \subseteq V$ (H is a subset of V). Then H is a <u>subspace</u> of V if (i) $0 \in H$

addition from V

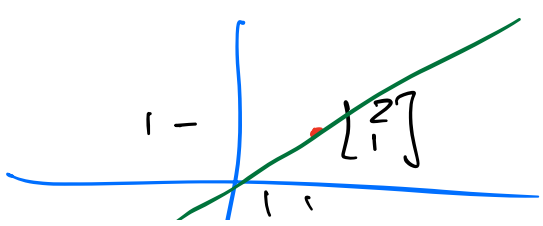
(ii) If $u, v \in H$ then $u+v \in H$

(iii) If $u \in H$, $c \in \mathbb{R}$ then $cu \in H$

scalar mult. from V

Theorem H is a subspace of V exactly when $H \subseteq V$ and H is a vector space.

Examples



$V = \mathbb{R}^2$

Green line is a subspace of $\mathbb{R}^2$.

Any line through the origin is a subspace.

The set containing the origin $\{0\} = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$.

Clearly $0+0=0$ and $c0=0$ $\forall c \in \mathbb{R}$

so $\{0\}$ is a subspace.

The zero subspace is a subspace of $\mathbb{R}^2$.

Finally, $\mathbb{R}^2$ is a subspace by defn.